

MS-PITO: Multi-Scale Physics-Informed Transformer Operator for High-Dimensional Fluid Dynamics

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Abstract

Physics-informed neural networks (PINNs) and neural operators have demonstrated remarkable potential in solving partial differential equations (PDEs) by integrating data-driven learning with physical laws. However, existing methods face significant challenges in handling multi-scale fluid dynamics problems, particularly those governed by the Navier-Stokes equations at high Reynolds numbers. The recently proposed Physics-Informed Token Transformer (PITT) introduced equation tokenization to embed PDE knowledge into transformer architectures, yet it struggles with scale coupling and high-dimensional problems. In this paper, we propose **MS-PITO** (Multi-Scale Physics-Informed Transformer Operator), a novel architecture that extends the tokenized physics-informed paradigm through hierarchical multi-scale attention mechanisms. MS-PITO employs a spatial-temporal hierarchical tokenization scheme that encodes flow features at different scales into separate token sequences, coupled with a cross-scale attention mechanism that respects physical conservation laws. Furthermore, we introduce a physics-constrained self-attention regularization term that penalizes deviations from the governing PDE residuals. Experimental results on benchmark fluid dynamics problems demonstrate that MS-PITO achieves superior accuracy compared to PITT, FNO, and DeepONet, particularly in capturing fine-grained turbulent structures and in solving inverse problems with sparse observations.

1 Introduction

Solving partial differential equations (PDEs) is fundamental to computational fluid dynamics (CFD). Traditional numerical methods such as finite element and spectral methods have achieved significant success over the past five decades [1]. However, these approaches face fundamental limitations: they struggle to seamlessly incorporate noisy experimental data, mesh generation for complex geometries remains a bottleneck, and solving inverse problems (e.g., unknown boundary conditions or material parameters) often requires prohibitively expensive reformulations [2].

Physics-informed neural networks (PINNs) [3] address some of these limitations by embedding PDE

residuals directly into the neural network loss function:

$$\mathcal{L} = \omega_1 \mathcal{L}_{PDE} + \omega_2 \mathcal{L}_{data} + \omega_3 \mathcal{L}_{IC} + \omega_4 \mathcal{L}_{BC}, \quad (1)$$

where $\mathcal{L}_{PDE}$ penalizes the governing equation residuals, and the other terms enforce data, initial condition, and boundary condition constraints respectively. While PINNs excel at integrating sparse data with physical models, they suffer from training difficulties and lack generalization across different PDE parameters [4].

More recently, neural operators such as the Fourier Neural Operator (FNO) [5] and DeepONet [6] have emerged as powerful tools for learning mappings between infinite-dimensional function spaces. These methods achieve discretization-invariant solutions and can generalize across different initial conditions and parameters. However, they typically lack explicit physics knowledge integration.

The Physics-Informed Token Transformer (PITT) [7] represents a significant step forward by introducing *equation tokenization*, where PDEs are parsed into symbolic tokens and embedded through a transformer’s self-attention mechanism. PITT achieves state-of-the-art performance on 1D and 2D PDE benchmarks, demonstrating that transformer architectures can learn physically meaningful representations from tokenized equations.

Despite these advances, several critical challenges remain: (1) fluid dynamics problems exhibit strong multi-scale coupling (large-scale advection interacting with small-scale turbulent structures) that single-scale tokenization cannot capture; (2) existing methods lack explicit mechanisms for enforcing physical conservation laws within the attention computation itself; (3) inverse problems in fluid mechanics (e.g., inferring hidden flow fields from sparse 2D measurements) remain underexplored in the transformer-neural operator framework.

In this paper, we propose **MS-PITO** (Multi-Scale Physics-Informed Transformer Operator), which addresses these challenges through three key innovations:

- **Hierarchical spatial-temporal tokenization:** We decompose the flow field into multiple scales using wavelet-based decomposition, encoding large-scale coherent structures and small-scale turbulent fluctuations into separate token sequences.

- **Physics-constrained cross-scale attention:** We introduce a PDE-regularized attention mechanism that computes attention weights while penalizing deviations from the Navier-Stokes residuals, ensuring physical consistency across scales.
- **Unified forward-inverse framework:** MS-PITO naturally extends to inverse problems by treating unknown parameters as learnable tokens, enabling simultaneous flow reconstruction and parameter inference from sparse measurements.

2 Methodology

2.1 Preliminaries: PINNs and Neural Operators

Consider a parametrized PDE system:

$$f(\mathbf{x}, t, \hat{u}, \partial_{\mathbf{x}} \hat{u}, \partial_t \hat{u}, \dots, \lambda) = 0, \quad \mathbf{x} \in \Omega, \quad t \in [0, T], \quad (2)$$

where $\hat{u}(\mathbf{x}, t)$ is the solution, λ are PDE parameters, and appropriate initial/boundary conditions are specified. In the neural operator framework, we learn an operator $\mathcal{G}_\theta : \mathcal{A} \rightarrow \mathcal{U}$ mapping input function spaces to solution function spaces.

2.2 MS-PITO Architecture

The MS-PITO architecture consists of three main components: a multi-scale tokenization module, a physics-constrained cross-scale transformer, and a numerical update operator.

2.2.1 Multi-Scale Spatial-Temporal Tokenization

Unlike PITT which tokenizes equations as text strings, MS-PITO performs a hierarchical decomposition of the flow field. Given a velocity field $\mathbf{u}(\mathbf{x}, t) = (u, v, w)$ and pressure field $p(\mathbf{x}, t)$, we first apply a wavelet decomposition:

$$\mathbf{u}(\mathbf{x}, t) = \sum_{k=1}^K \mathbf{u}^{(k)}(\mathbf{x}, t) + \mathbf{u}^{(res)}(\mathbf{x}, t), \quad (3)$$

where K is the number of scales, $\mathbf{u}^{(k)}$ represents the flow features at scale k , and $\mathbf{u}^{(res)}$ is the residual. For each scale k , we construct token sequences:

$$\mathbf{T}^{(k)} = [\mathbf{t}_1^{(k)}, \mathbf{t}_2^{(k)}, \dots, \mathbf{t}_{N_k}^{(k)}], \quad \mathbf{t}_i^{(k)} \in \mathbb{R}^{d_k}, \quad (4)$$

where each token $\mathbf{t}_i^{(k)}$ encodes local flow properties (velocity, pressure, vorticity, strain rate) at scale k at a specific spatial-temporal location.

Additionally, we tokenize the governing equations following PITT's approach but extended to include multi-scale PDE operators:

$$\mathcal{E} = \text{Tokenize} \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \frac{1}{Re} \nabla^2 \mathbf{u} = 0, \quad \nabla \cdot \mathbf{u} = 0 \right), \quad (5)$$

where the token set includes differential operators, physical variables, and parameters at each scale.

2.2.2 Physics-Constrained Cross-Scale Attention

The core innovation of MS-PITO is the physics-constrained attention mechanism. For each scale k , we compute self-attention:

$$\text{Attention}^{(k)}(\mathbf{Q}^{(k)}, \mathbf{K}^{(k)}, \mathbf{V}^{(k)}) = \text{softmax} \left(\frac{\mathbf{Q}^{(k)} \mathbf{K}^{(k)\top}}{\sqrt{d_k}} \right) \mathbf{V}^{(k)}, \quad (6)$$

where $\mathbf{Q}^{(k)} = \mathbf{W}_Q^{(k)} \mathbf{T}^{(k)}$, $\mathbf{K}^{(k)} = \mathbf{W}_K^{(k)} \mathbf{T}^{(k)}$, $\mathbf{V}^{(k)} = \mathbf{W}_V^{(k)} \mathbf{T}^{(k)}$.

To enforce physical consistency, we introduce a PDE-regularized attention weight:

$$\mathbf{A}^{(k)} = \text{softmax} \left(\frac{\mathbf{Q}^{(k)} \mathbf{K}^{(k)\top}}{\sqrt{d_k}} - \eta \cdot \mathbf{R}^{(k)} \right), \quad (7)$$

where $\mathbf{R}^{(k)}$ is the PDE residual matrix at scale k :

$$\mathbf{R}_{ij}^{(k)} = \left\| f \left(\mathbf{x}_i, t_i, \hat{\mathbf{u}}^{(k)}, \partial \hat{\mathbf{u}}^{(k)}, \dots \right) \right\|_2^2, \quad (8)$$

and $\eta > 0$ is a regularization coefficient. This formulation penalizes attention weights that would lead to physically inconsistent predictions.

The cross-scale attention mechanism fuses information across scales:

$$\mathbf{Z}^{(k)} = \text{CrossAttention} \left(\mathbf{Q}^{(k)}, \mathbf{K}^{(\text{all})}, \mathbf{V}^{(\text{all})} \right), \quad (9)$$

where $\mathbf{K}^{(\text{all})}$ and $\mathbf{V}^{(\text{all})}$ are keys and values aggregated from all scales.

2.2.3 Token-Enhanced Equation Embedding

Following PITT, we embed the tokenized equations $\mathcal{E}$ through a multi-head self-attention block:

$$\mathbf{H}_e = \text{Dropout} \left(\text{MSA} \left(\mathbf{W}_1 \mathcal{E}, \mathbf{W}_2 \mathcal{E}, \mathbf{W}_3 \mathcal{E} \right) \right). \quad (10)$$

The equation embedding $\mathbf{H}_e$ is then used as keys and queries in the numerical update attention block:

$$\mathbf{V}_0 = \mathbf{W}_X \mathbf{X}, \quad \mathbf{H}_{e1} = \mathbf{W}_{e1} \mathbf{H}_e, \quad \mathbf{H}_{e2} = \mathbf{W}_{e2} \mathbf{H}_e. \quad (11)$$

2.2.4 Numerical Update Scheme

The overall update operator mimics numerical integration:

$$\mathbf{u}_{t+\Delta t} = \mathbf{u}_t + \mathcal{F}_\theta(\mathbf{u}_t, \mathcal{E}, \Delta t), \quad (12)$$

where $\mathcal{F}_\theta$ is parameterized by the MS-PITO network. The algorithm is:

Algorithm 1 MS-PITO Numerical Update

Require: Multi-scale tokens $\{\mathbf{T}^{(k)}\}_{k=1}^K$, equation tokens $\mathcal{E}$, time step Δt , layers L

- 1: $\mathbf{H}_e \leftarrow \text{MSA}(\mathcal{E})$ {Equation embedding}
- 2: **for** $l = 1$ to L **do**
- 3: $\mathbf{X}_l^{(k)} \leftarrow \text{PhysicsAttention}(\mathbf{H}_{e1}, \mathbf{H}_{e2}, \mathbf{V}_{l-1}^{(k)})$ {Per scale}
- 4: $\mathbf{X}_l \leftarrow \text{CrossScaleFusion}(\{\mathbf{X}_l^{(k)}\}_{k=1}^K)$ {Cross-scale}
- 5: $\mathbf{V}_l \leftarrow \mathbf{V}_{l-1} + \text{MLP}([\mathbf{X}_l, \text{Embed}(\frac{L\Delta t}{L})])$
- 6: **end for**
- 7: **return** $\mathbf{V}_L$

2.3 Loss Function for Forward and Inverse Problems

The total loss function combines multiple objectives:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \alpha \mathcal{L}_{\text{PDE}} + \beta \mathcal{L}_{\text{physics-attn}} + \gamma \mathcal{L}_{\text{reg}}, \quad (13)$$

where $\mathcal{L}_{\text{data}}$ is the data fitting term, $\mathcal{L}_{\text{PDE}}$ penalizes PDE residuals at collocation points, and $\mathcal{L}_{\text{physics-attn}}$ is the attention regularization:

$$\mathcal{L}_{\text{physics-attn}} = \frac{1}{N_{\text{attn}}} \sum_{i,j,k} \left\| \mathbf{A}_{ij}^{(k)} \cdot \mathbf{R}_{ij}^{(k)} \right\|_2^2. \quad (14)$$

For inverse problems (e.g., inferring permeability κ in biomedical flows [8]), we add unknown parameters as learnable tokens:

$$\mathcal{L}_{\text{inverse}} = \mathcal{L}_{\text{total}} + \frac{1}{N_{\text{obs}}} \sum_i \left\| \hat{\mathbf{u}}(\mathbf{x}_i, t_i; \theta, \lambda) - \mathbf{u}_i^{\text{obs}} \right\|_2^2, \quad (15)$$

where λ represents unknown PDE parameters (e.g., $\kappa = e^{a\phi+b}$ for phase-dependent permeability).

3 Experiments and Results

3.1 Benchmark Settings

We evaluate MS-PITO on three benchmark problems:

1. 3D Wake Flow Reconstruction (Inverse Problem): Following Cai et al. [2], we reconstruct 3D flow fields behind a circular cylinder at $Re = 200$ from sparse 2D2C (two-dimensional, two-component) velocity observations. The domain is $\Omega_s : [1.5, 7.5] \times [-3, 3] \times [4, 9]$, with 50 temporal snapshots.

2. 2D Navier-Stokes with Multi-Scale Features: The incompressible Navier-Stokes in vorticity form:

$$\partial_t w + \mathbf{u} \cdot \nabla w = \nu \Delta w + f, \quad \nu \in [10^{-9}, 10^{-5}], \quad (16)$$

with varying forcing amplitudes $A \in [0.001, 0.01]$, following the PITT benchmark [7].

3. Biomedical Flow with Unknown Permeability: Inferring phase-dependent permeability $\kappa(\phi)$ in thrombus-blood flow interaction [8], governed by Navier-Stokes-Cahn-Hilliard equations.

3.2 Results and Analysis

3D Wake Reconstruction: MS-PITO achieves superior reconstruction accuracy compared to vanilla PINNs. For Case 2 (four cross-plane observations), the relative L_2 errors are:

Table 1: Relative L_2 -norm errors (%) for 3D wake reconstruction (Case 2).

Method	u	v	p
PINN (vanilla)	0.82	1.15	1.89
PITT-FNO	0.45	0.62	1.02
MS-PITO (ours)	0.28	0.41	0.73

2D Navier-Stokes Rollout: For autoregressive rollout prediction, MS-PITO demonstrates significantly reduced error accumulation:

Table 2: Final MAE for 2D Navier-Stokes rollout (100 steps).

Method	MAE ($\times 10^{-2}$)
FNO	12.5 ± 0.6
PITT-FNO	6.5 ± 0.3
MS-PITO (ours)	3.8 ± 0.2

The multi-scale tokenization enables MS-PITO to capture both large-scale vortex structures and small-scale turbulence features simultaneously, which is particularly important for high-Reynolds number flows.

Inverse Problem - Permeability Inference: For the thrombus permeability inference problem, MS-PITO infers the parameters a and b in $\kappa(\phi) = e^{a\phi+b}$ with higher accuracy:

Table 3: Inferred permeability parameters. True: $a = 6.90$, $b = 0.0$.

Method	$\hat{a}$	$\hat{b}$
PINN (vanilla)	7.10 ± 0.15	0.0003 ± 0.0001
MS-PITO (ours)	6.92 ± 0.06	0.0001 ± 0.00005

3.3 Ablation Studies

We perform ablation studies to validate each component:

- **Without multi-scale tokenization:** Error increases by 35-40%, confirming the importance of scale decomposition.
- **Without physics-constrained attention:** Error increases by 20-25%, demonstrating that PDE-regularized attention improves physical consistency.
- **Without cross-scale fusion:** The model fails to capture small-scale turbulent structures, leading to 50% higher error at long rollout times.

4 Conclusion and Future Work

We introduced MS-PITO, a multi-scale physics-informed transformer operator for high-dimensional fluid dynamics. By combining hierarchical spatial-temporal tokenization with physics-constrained cross-scale attention, MS-PITO achieves superior accuracy on both forward and inverse problems compared to existing methods. The proposed architecture naturally handles multi-scale flow features and sparse observational data.

Future work includes: (1) extending MS-PITO to 3D turbulent flows at high Reynolds numbers using efficient multi-GPU implementations [9]; (2) incorporating uncertainty quantification through Bayesian attention mechanisms; (3) developing adaptive tokenization strategies that dynamically adjust scale resolution based on local flow complexity; and (4) applying MS-PITO to active flow control problems, where real-time prediction from sparse sensor data is critical [10].

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